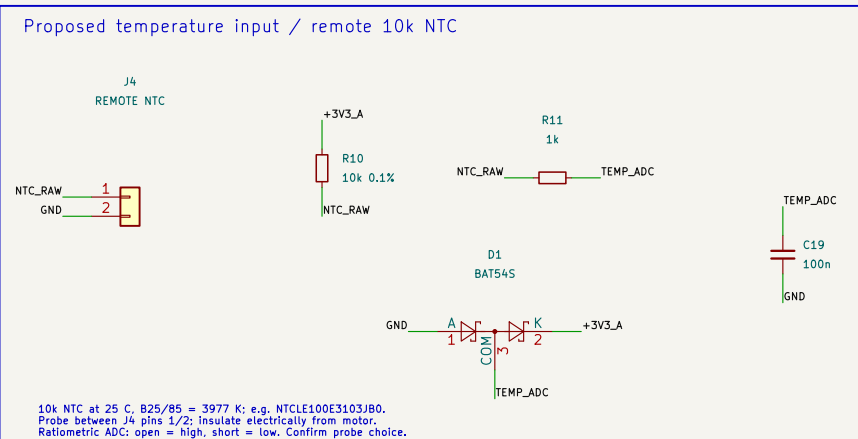
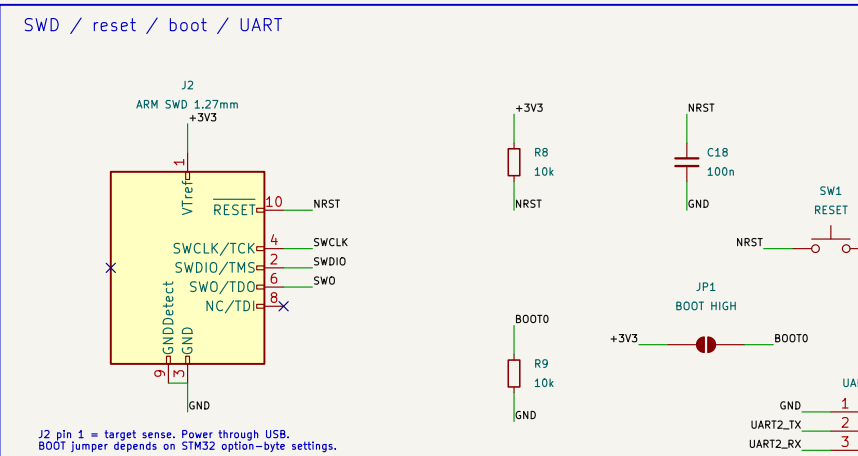
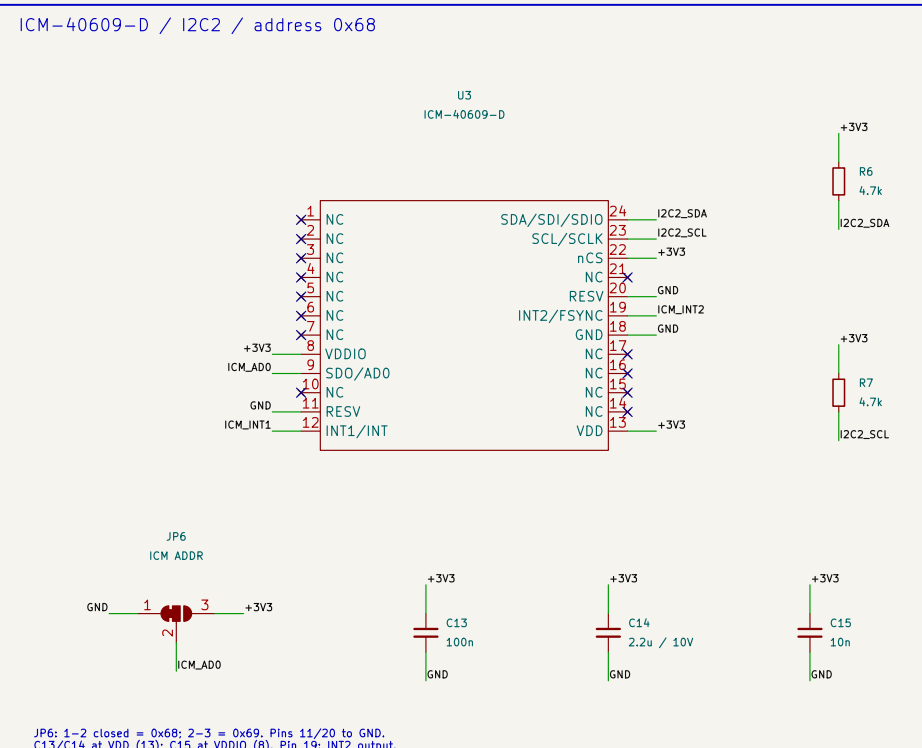
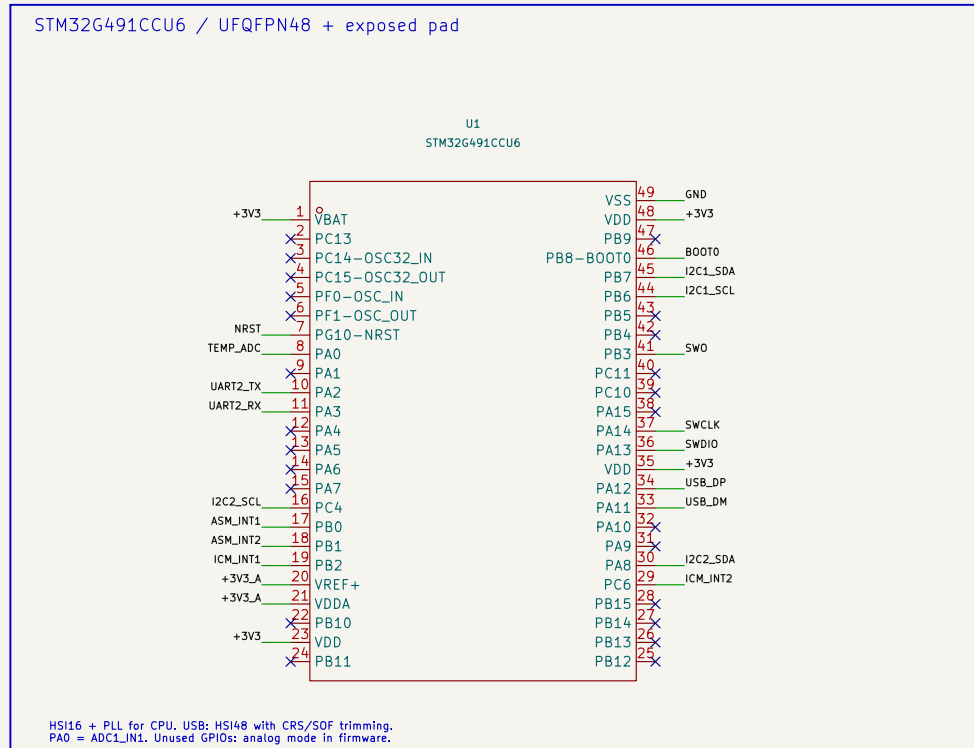
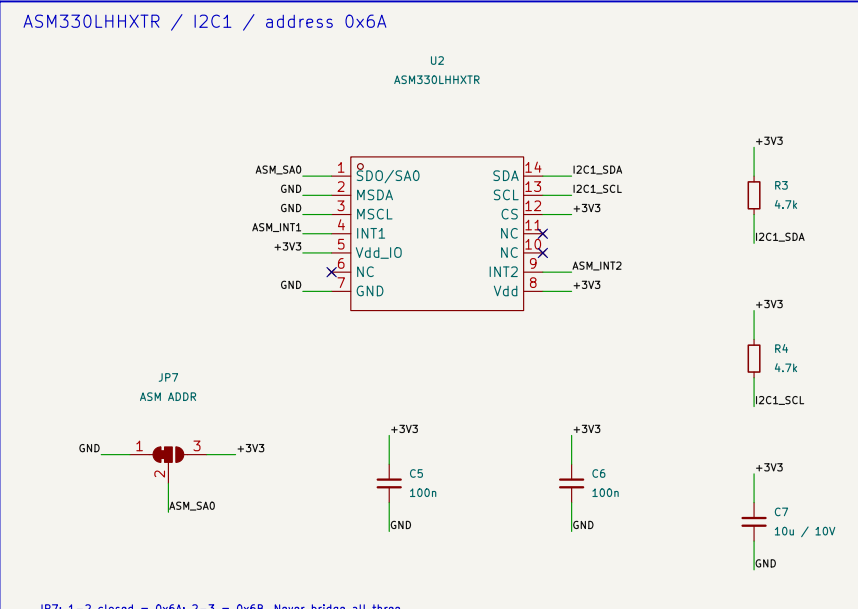
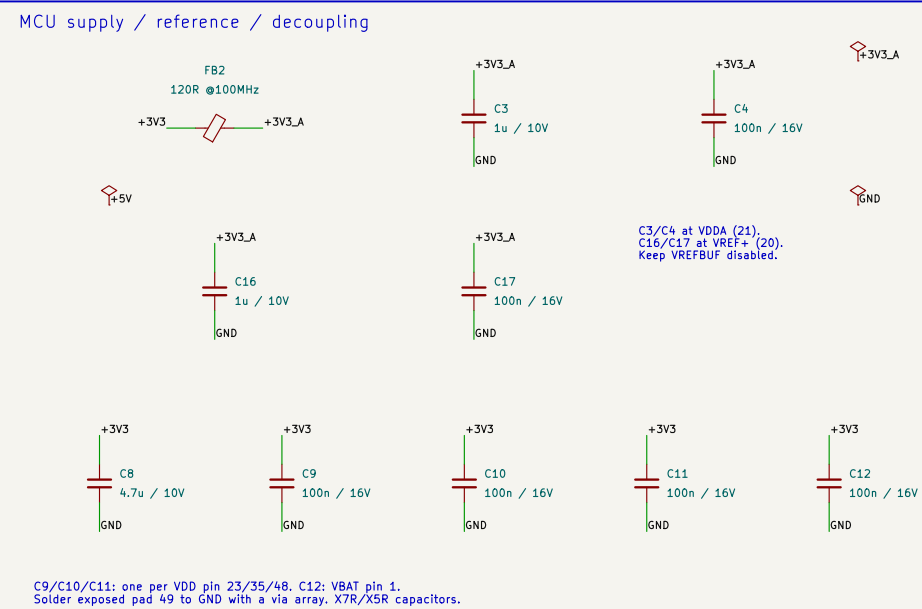
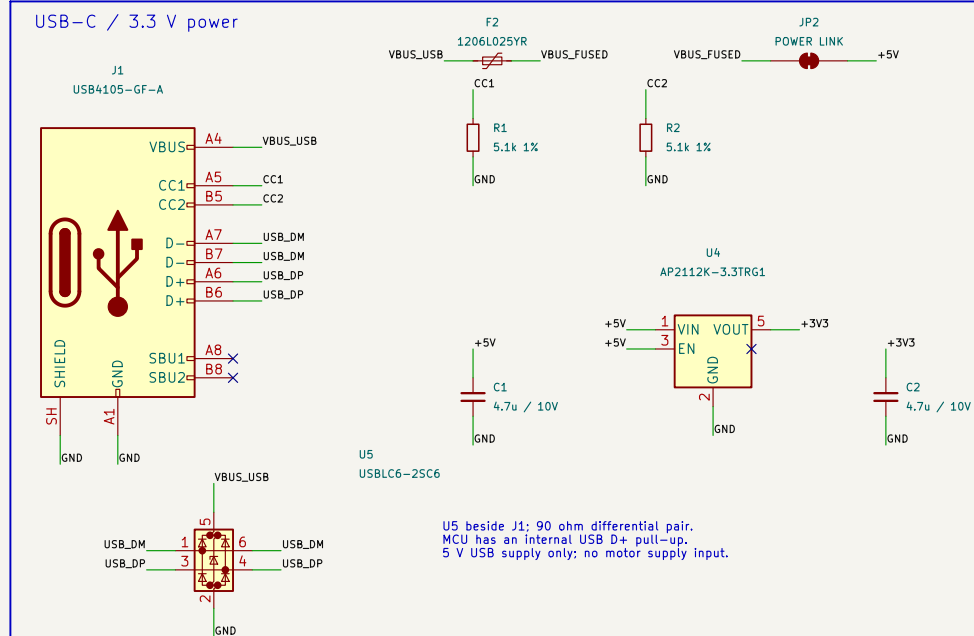




- ## Implementation notes / bring-up assumptions
1. Preserve separate I2C buses: PB6/PB7 = I2C1 AF4; PC4/PAB = I2C2 AF4. Start at 400 kHz.
 2. 4.7k pull-ups assume short PCB traces and ≈ 75 pF bus capacitance. Verify rise time on hardware.
 3. Begin at < 4 ksample per IMU with FIFO buffers. Full 12 kHz acquisition requires an SPI readout.
 4. IMU interrupts: PB0/PB1 = ASM INT1/INT2; PB2/PBC = ICM INT1/INT2. Configure U3 pin 19 as INT2.
 5. Keep IMUs rigidly coupled to the motor housing, away from board flex, fasteners, and regulator heat.
 6. USB crystal-less operation is specified for -15 to $+85$ C. HS148 requires CRS synchronization.
 7. AP2121 replaces AMS1117 to support ceramic capacitors and improve dropout headroom.
 8. NTC is a proposed circuit, not a confirmed sensor requirement. Use calibrated vendor R/T data.
 9. Verify USB input buffer, suspend, current, regulator temperature, and probe noise on the prototype.
 10. Schematics only: PCB placement/routing and fabrication review remain. See DESIGN_Notes.txt.